Practical - 3

Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

A. Theory & Analysis

1. How does the layout adapt for mobile, tablet, and desktop screens?

The layout follows a mobile-first approach. On mobile screens (375px), navigation collapses into a hamburger menu and multi-column card grids stack vertically in single columns to avoid horizontal scrolling. On medium tablet screens (768px), cards display in a 2-column layout. On desktop viewports (≥992px), cards expand into a 4-column row with balanced spacing.

2. Which layout technique is used and why?

I used both Flexbox and CSS Grid: (1) Flexbox: Bootstrap 5's standard grid (row, col-*) and utility classes (d-flex, justify-content-between, align-items-center) are built on CSS Flexbox, which I used for one-dimensional component alignment, navbar items, and responsive card wrapping; (2) CSS Grid: Used for two-dimensional page layouts where both columns and rows are defined simultaneously, such as .portal-grid with grid-template-columns: repeat(auto-fit, minmax(220px, 1fr)) for fluid card arrangements.

3. Are typography, spacing, colors, and contrast readable?

Yes. Clean system fonts (system-ui, Segoe UI, Roboto) are used with comfortable line heights. The color palette centers on a university primary blue theme (#0d47a1 and #0d6efd) with high contrast against white and light-gray backgrounds for clear readability. Spacing is maintained using consistent modular utility classes (p-*, m-*, g-*).

4. Is the framework used consistently without unnecessary inline CSS?

Yes. Bootstrap 5 classes are used consistently for components, layout containers, and utilities. Custom styles are separated into external stylesheets (such as css/dashboard.css and css/register.css) rather than cluttered inline styles, preserving clean separation of concerns.

B. Implementation Details

I developed responsive layouts across the portal using Bootstrap 5.3.3 and custom CSS. I organized the dashboard attendance cards using a unified primary-blue style with consistent borders and badges. I tested the responsiveness across desktop (1280px), tablet (768px), and mobile (375px) viewports to verify that navbar elements wrap properly and cards stack cleanly without horizontal clipping.

C. Evidence of Work

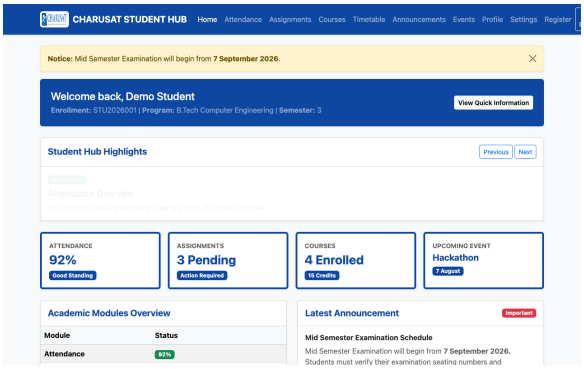


Figure 1: Desktop Layout (1280px) with 4-Column Grid Alignment

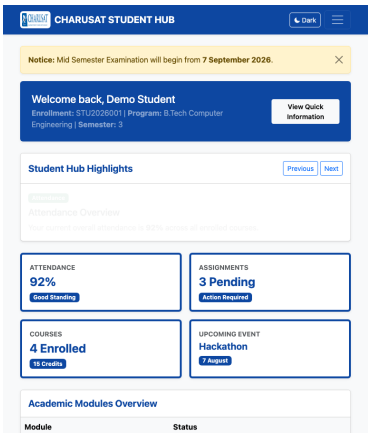


Fig 2: Tablet Portrait Layout (768px) with 2-Column Responsive Stacking

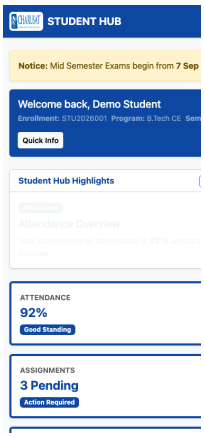


Fig 3: Mobile Viewport Layout (375px) with Fluid Single-Column Flow

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned how to build mobile-first responsive web layouts using Bootstrap Flexbox grids, CSS Grid auto-fit properties, and responsive media query breakpoints across desktop, tablet, and mobile devices.